

To what extent is the rate of adoption of AI at the workplace associated with the increase in the unemployment rate? A comparison of youth and the total labour force

Tutorial 6 Group 2
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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

Artificial Intelligence (AI) is changing the way how firms hire workers, manage and organise work and use technology which has led to an increasingly important social and economic issue. With recent developments in generative AI, especially after the release of ChatGPT, in November of 2022, AI tools are more visible and widely used across different industries. It is argued that AI in the workplace increases productivity but it also may reduce demand of some type of labour, especially in entry level jobs (Cazzaniga et al., 2024).

This becomes increasingly relevant for young workers aged 18-24. As workers in this age group, often have low to nil level of work experience and are very likely to work in temporary, entry level jobs. Therefore, this results in these workers being vulnerable when employers change the skills that they demand or when certain tasks can be fulfilled by AI. Youth Employment is therefore a social problem, because unemployment in early stages of their career/life can negatively affect future earnings, skill development and work experience which is required for higher skilled jobs (International Labour Organisation, 2024).

This project examines whether the rise of national AI-Development capacity is linked and associated with worse labour outcomes for young workers compared to the total labour force. We compare data from countries from the years 2021-2025. The aim is whether there is an observable association between AI adoption and youth unemployment relative to overall unemployment.

Part 2 - Data Sourcing

2.1 Description and Limitations

The data was compiled from the following datasets:

The World Bank Data:

- Unemployment, youth total (% of total labor force ages 15-24) (modeled ILO estimate): `SL.UEM.1524.ZS`
- Unemployment, total (% of total labor force) (modeled ILO estimate): `SL.UEM.TOTL.ZS`

Eurostat:

- AI adoption rate, or AI used in enterprises: `isoc_eb_ai`

Table 1: Datasets used in this report

Dataset	Unit of observation	Key variables	Source	Period
World Bank (WDI)	Country-year	total_unemp, youth_unemp (% of labour force)	World Bank / ILO modelled estimates (ILOSTAT)	1991-2024 (annual)
Eurostat (isoc_eb_ai)	Country-year	values (% of enterprises using AI)	Eurostat: ICT usage in enterprises survey	2021, 2023, 2024, 2025

Limitations The Eurostat dataset of AI adoption rate only consists of European countries with temporal data from 2021 to 2025. Because of this, the findings cannot be generalised to other regions and the whole world. A wider dataset with more temporal data points can help increase accuracy of the findings and produce more predictive outcomes, but since the wave of AI usage is a fairly new societal shift, it is reasonable that there are currently few temporal data to observe. Another point worth mentioning is that AI adoption rate in 2022 is unavailable from the source, so we lack one important temporal value of the year ChatGPT was launched. To accommodate this, this paper looks at any difference of the association of AI usage and unemployment rate before and after the launch of ChatGPT, which works as the main event that signals the rise of the AI wave.

2.2 Loading the data

Table 2: Size of each raw dataset

Dataset	Rows	Columns
wb_raw (World Bank)	1156	14
ai_raw (Eurostat)	72527	8

2.3 Describe the type of variables included

Unemployment rate - World Bank

Two indicators are used in this report: total unemployment (SL.UEM.TOTL.ZS) and youth unemployment, ages 15-24 (SL.UEM.1524.ZS). Both are expressed as a percentage of the labour force. Coverage of the raw data is worldwide and annual, during 1991-2024.

AI use in enterprises - Eurostat

Out of all indicators provided in (isoc_eb_ai), we selected the percentage of enterprises with 10 or more employees that use at least one AI technology including speech recognition, machine learning, etc. (E_AI_TANY) The dataset covers mainly data of EU countries for the years 2021, 2023, 2024 and 2025, prompting the need to extract only the unemployment rate of youths and the total population in EU countries from the World Bank datasets.

Part 3 - Quantifying

3.1 Data cleaning

As we have chosen to focus on the years 2021-2024 for better comparison of the impact of AI before and after the launch of ChatGPT in 2022, we need to filter data of unemployment within this period (OpenAI, 2022).

Out of many dimensions available in the raw data source, we filtered for number of enterprises using at least one of the AI technologies.

Table 3: Variables in the merged (combined) dataset

Variable	Type	Meaning
country	character	Country name
iso2c	character	2-letter country code
year	integer	Year of observation
total_unemp	numeric	Total unemployment (% of labour force)

Variable	Type	Meaning
youth_unemp	numeric	Youth unemployment, ages 15-24 (% of labour force)
values	numeric	Enterprises using at least one AI technology (%)
ai_group	factor	AI adoption group: Low / Medium / High (new variable)
youth_ratio	numeric	Youth-to-total unemployment ratio (new variable)

3.2 Generate necessary variables

Variable 1: AI adoption rate, or AI usage intensity

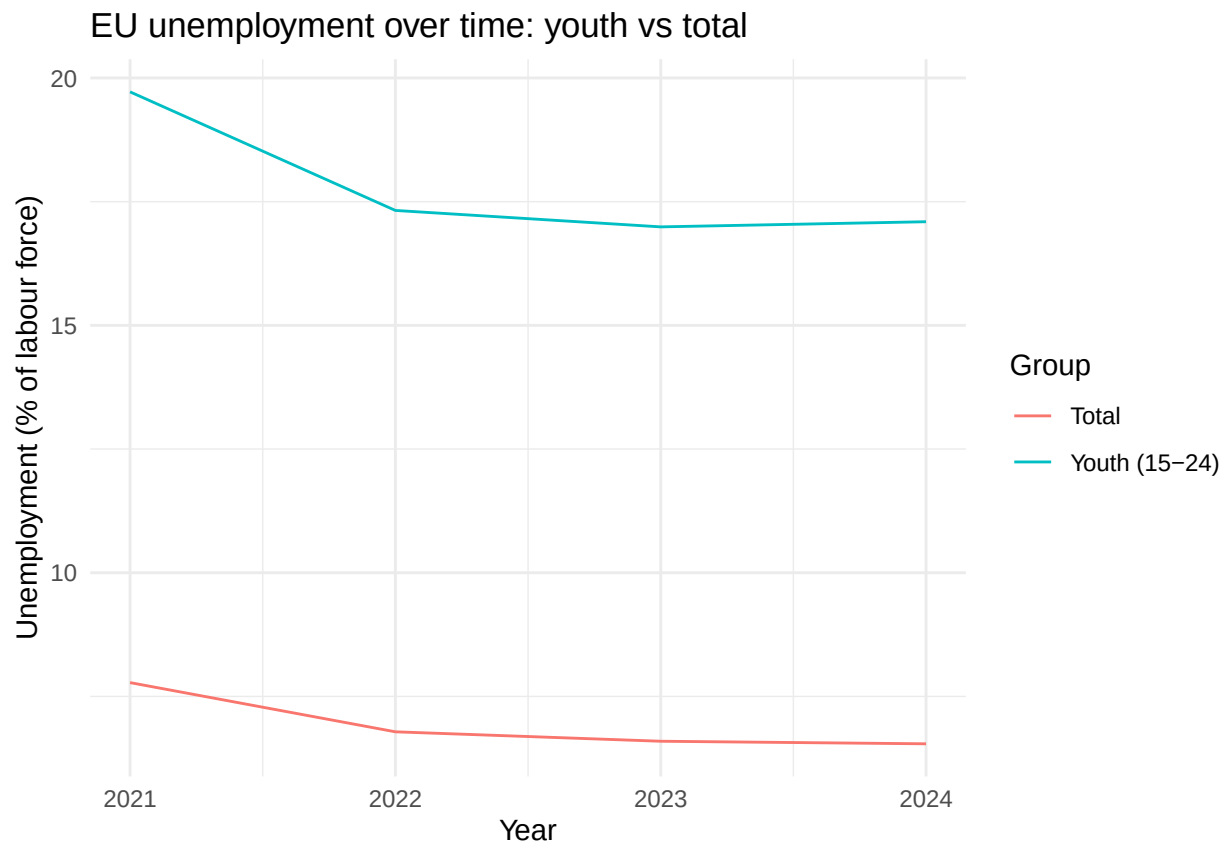
```
combined <- wb_clean %>%
  left_join(ai_clean, by = c("iso2c" = "geo", "year" = "TIME_PERIOD")) %>%
  mutate(ai_group = case_when(
    values < 10 ~ "Low",
    values < 30 ~ "Medium",
    TRUE      ~ "High"
  ))
```

Variable 2:

```
wb_clean <- wb_clean %>%
  mutate(youth_ratio = youth_unemp / total_unemp)
combined <- combined %>%
  mutate(youth_ratio = youth_unemp / total_unemp)
```

```
## # A tibble: 3 x 4
##   ai_group      n mean_total_unemp mean_youth_ratio
##   <chr>    <int>         <dbl>         <dbl>
## 1 High      169           7.12           2.57
## 2 Low       500           7.29           2.76
## 3 Medium   373           5.76           2.70
```

3.3 Visualize temporal variation

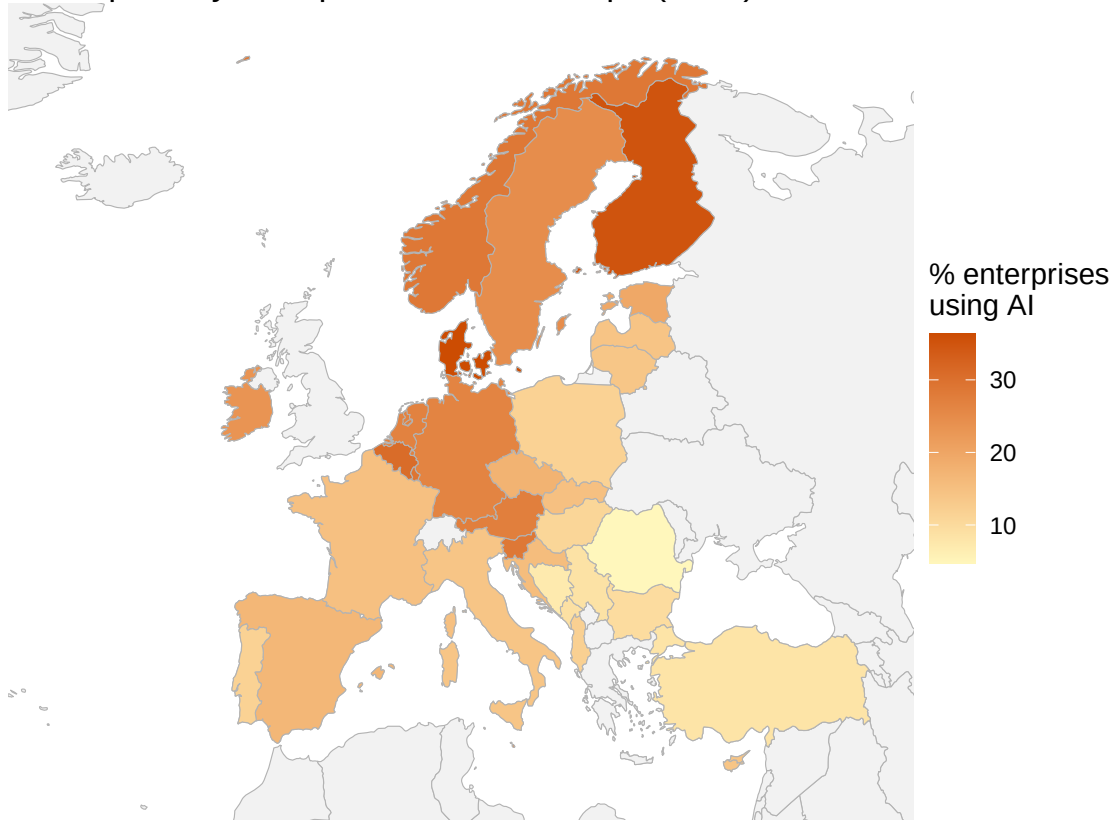


ai_group - bins the continuous values into Low, Medium, and High, so we can compare youth unemployment outcomes across different levels of AI adoption.

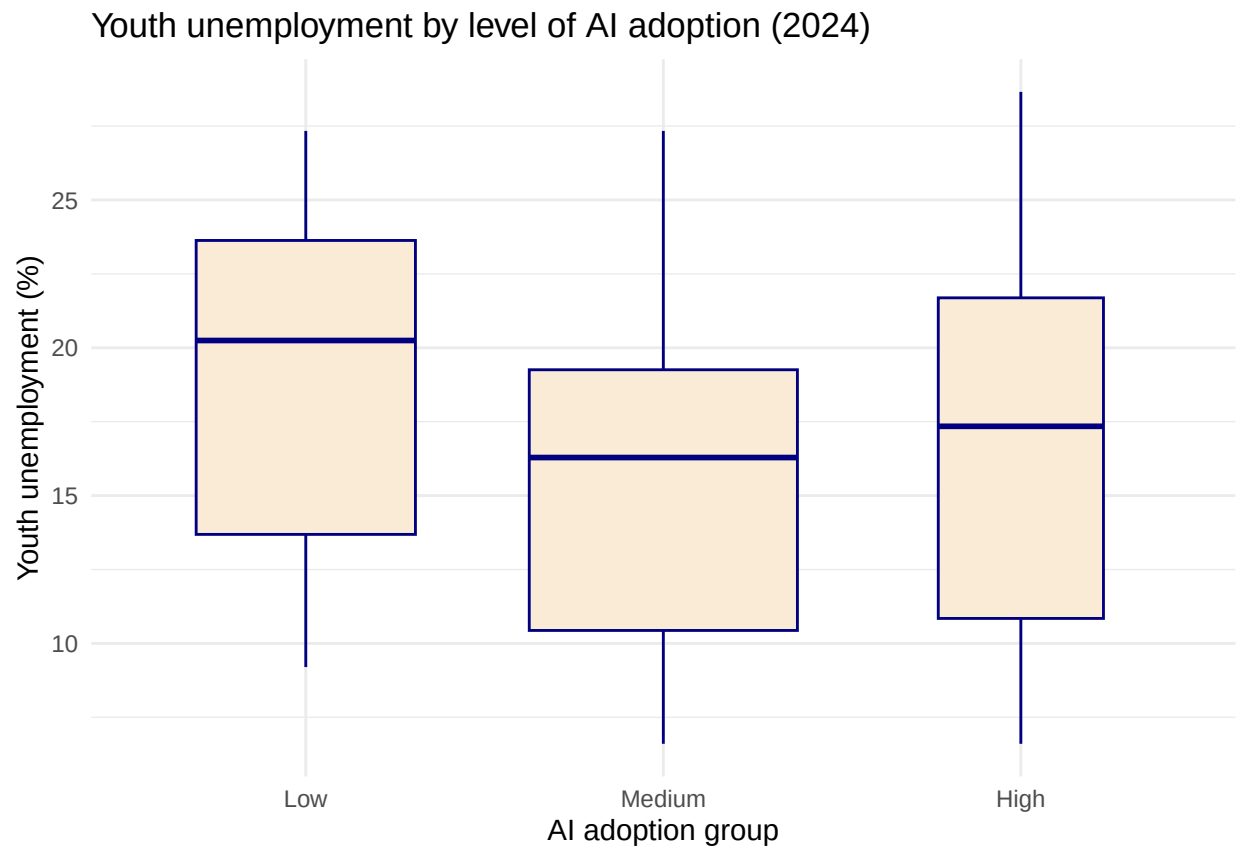
3.4 Visualize spatial variation

```
##  
## Attaching package: 'maps'  
  
## The following object is masked from 'package:purrr':  
##  
## map
```

AI adoption by enterprises across Europe (2024)



3.5 Visualize sub-population variation

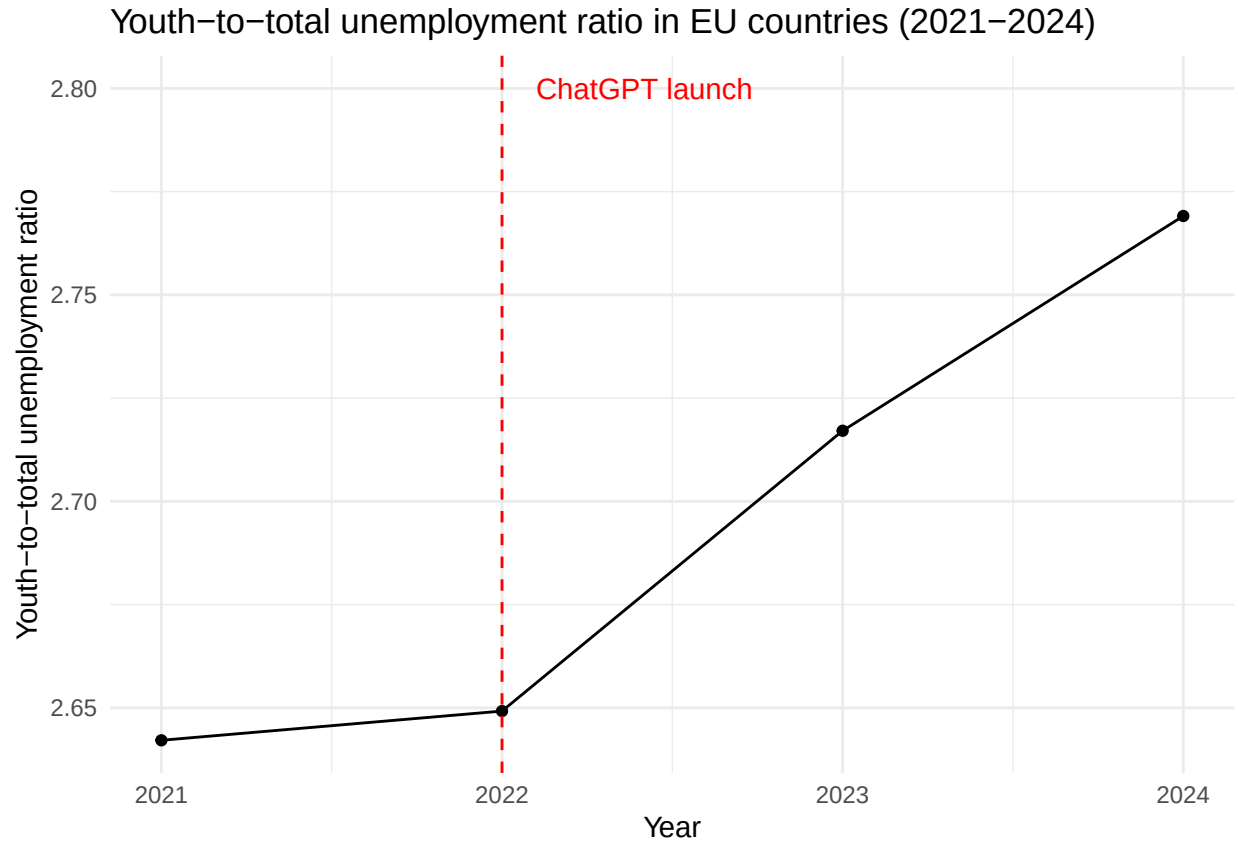


Here you provide a description of why the plot above is relevant to your specific social problem. [tbu]

3.6 Event analysis

Analyze the relationship between two variables.

```
## # A tibble: 4 x 2
##   year mean_youth_ratio
##   <dbl>         <dbl>
## 1  2021           2.64
## 2  2022           2.65
## 3  2023           2.72
## 4  2024           2.77
```



The graph shows the average youth-to-total unemployment ratio across EU countries from 2021 to 2024. The ratio rises steadily after 2022 - the year ChatGPT launched - suggesting that young workers may be increasingly disadvantaged relative to the broader workforce as AI adoption grows. However, this is an association, not proof: other factors such as post-COVID labor market shifts or broader economic conditions could equally explain the trend.

Part 4 - Discussion

4.1 Discuss your findings

AI Adoption and Unemployment in European Countries

1. Youth unemployment is consistently much higher than overall unemployment

From 2021-2024, youth unemployment (ages 15-24) remained substantially above total unemployment in all years. The youth-to-total unemployment ratio ranged from 2.64 to 2.77, meaning young people were approximately 2.6-2.8 times more likely to be unemployed than the average worker.

2. The youth unemployment gap appears to be increasing

The youth-to-total unemployment ratio increased from 2.64 in 2021 to 2.77 in 2024. This suggests that young workers may have become increasingly disadvantaged relative to the broader workforce during the period in which AI adoption accelerated. However, this is an association and does not prove causation.

3. Countries with medium AI adoption have the lowest unemployment

The analysis grouped countries into Low, Medium, and High AI adoption categories. Countries in the

Medium AI adoption group had the lowest average total unemployment (5.76%) and youth unemployment (15.3%).

4. High AI adoption is not associated with the worst youth outcomes

The Low AI adoption group had the highest average youth unemployment rate (19.1%), while the High AI adoption group averaged 17.6%. This suggests that higher AI adoption is not automatically associated with higher youth unemployment.

5. Youth unemployment remains disproportionately high regardless of AI level

The youth-to-total unemployment ratio remained relatively similar across AI groups (High: 2.57, Medium: 2.70, Low: 2.76), indicating that youth unemployment is a widespread challenge across Europe regardless of AI adoption intensity.

6. Country-specific differences appear important

The scatterplot and boxplot suggest substantial variation between countries within each AI adoption category. This indicates that national labour market conditions, economic structures, and policy differences may play a larger role than AI adoption alone.

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Cazzaniga, M., Jaumotte, F., Li, L., Melina, G., Panton, A. J., Pizzinelli, C., Rockall, E. J., Tavares, M. M., & Yao, K. (2024)

Eurostat (2026). Artificial intelligence by size class of enterprise [Data set]. https://doi.org/10.2908/ISOC_EB_AI

Global Employment Trends for Youth 2024. (2026). International Labour Organization. <https://www.ilo.org/publications/major-publications/global-employment-trends-youth-2024>

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World Bank. (2026). Unemployment, youth total (% of total labor force ages 15-24) (modeled ILO estimate) [Data set]. ILOSTAT. <https://data.worldbank.org/indicator/SL.UEM.1524.ZS>